

Estimate the quantity of objects in collections and make estimates when solving problems to determine the reasonableness of calculations

Learning intention

- estimate the quantity of objects in collections and make estimates when solving problems to determine the reasonableness of calculations.

Teach and model

- estimating the number of people in a large gathering.
- choosing which place value they would estimate to for different situations.
- checking the reasonableness of an addition calculation by using two- and three-digit numbers to the nearest 10 or hundred to estimate.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.